Introduction to GLMs

Contents

1 Introduction **2**
Configuring R . . . 2

2 Welcome **3**

3 What you should already know **3**
3.0.1 Epi 202: probability models . . . 3
3.0.2 Epi 203: inference for one or several homogenous populations . . . 3
3.0.3 Stat 108: linear regression models . . . 4

4 What we will cover in this course **4**

5 The scientific method and models **5**

6 Motivations for regression models **5**
6.1 Uses of regression models . . . 5
6.1.1 Relating the two lists . . . 6
6.1.2 Prediction, Inference, and Causal Inference . . . 7
6.2 Example: Adelie penguins . . . 8
6.3 Linear regression . . . 8
6.4 Curved regression lines . . . 9
6.5 Multiple regression . . . 10
6.6 Modeling non-Gaussian outcomes . . . 12
6.7 Why don't we use linear regression? . . . 13
6.8 Zoom out . . . 14
6.9 log transformation of dose? . . . 15
6.10 Logistic regression . . . 16

7 Structure of regression models **16**

8 Why link functions? **18**

9 Canonical links **19**

10 The model fitting process **20**
10.1 Step 0: Exploratory data analysis . . . 20
10.2 Step 1: Model specification . . . 21
10.3 Step 2: Parameter estimation . . . 22
10.4 Step 3: Checking model adequacy . . . 24
10.5 Step 4: Inference . . . 25

11 Choosing a model **26**

Exercises **28**

References **30**

1 Introduction

Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# library(printr) # inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
library(broom) # Summarizes key information about statistical objects in tidy tibbles
library(broom.helpers) # Provides suite of functions to work with regression model 'broom::tidy()' t
```

Here are some R settings I use in this document:

```
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
        # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")))

knitr::opts_chunk$set(message = FALSE)
options('digits' = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
conflicts_prefer(dplyr::filter) # use the `filter()` function from dplyr() by default
legend_text_size = 9
run_graphs = TRUE
```

2 Welcome

Welcome to Epidemiology 204: Quantitative Epidemiology III (Statistical Models).

Epi 204 is a course on **regression modeling**.

3 What you should already know

> ⚠ Warning
>
> Epi 202, Epi 203, and Sta 108 are prerequisites for this course. If you haven't passed one of these courses, talk to me ASAP.

3.0.1 Epi 202: probability models

- Probability distributions
  - binomial
  - Poisson
  - Gaussian
  - exponential

- Characteristics of probability distributions
  - Mean, median, mode, quantiles
  - Variance, standard deviation, overdispersion

- Characteristics of samples
  - independence, dependence, covariance, correlation
  - ranks, order statistics
  - identical vs nonidentical distribution (homogeneity vs heterogeneity)
  - Laws of Large Numbers
  - Central Limit Theorem for the mean of an iid sample

3.0.2 Epi 203: inference for one or several homogenous populations

- the maximum likelihood inference framework:
  - likelihood functions
  - log-likelihood functions
  - score functions
  - estimating equations
  - information matrices
  - point estimates
  - standard errors
  - confidence intervals
  - hypothesis tests
  - p-values

- Hypothesis tests for one, two, and >2 groups:
  - t-tests/ANOVA for Gaussian models
  - chi-square tests for binomial and Poisson models
  - nonparametric tests:
    - Wilcoxon signed-rank test for matched pairs
    - Mann–Whitney/Kruskal-Wallis rank sum test for ≥ 2 independent samples
    - Fisher's exact test for contingency tables
    - Cochran–Mantel–Haenszel-Cox log-rank test

For all of these prerequisite concepts and quantities, and especially for confidence intervals and p-values, you should know how **both**:

- how to compute them
- how to interpret them

3.0.3 Stat 108: linear regression models

- building models for Gaussian outcomes
  - multiple predictors
  - interactions
- regression diagnostics
- fundamentals of R programming; e.g.:
  - Wickham et al. (2023)
  - Dalgaard (2008)
- RMarkdown or Quarto for formatting homework[1]
  - LaTeX for writing math in RMarkdown/Quarto

4 What we will cover in this course

- Linear (Gaussian) regression models (review and more details)
- Regression models for non-Gaussian outcomes
  - binary
  - count
  - time to event
- Statistical analysis using R

We will start where Epi 203 left off: with linear regression models.

The type of the outcome variable Y largely determines which regression model we use (Table 1).

Table 1: Outcome types, examples, and the regression analyses that match them

Outcome	Examples	Analysis
binary	diseased/not; died/survived	logistic regression
count	number of infections per month; number of follow-up visits per patient	Poisson regression
time to event	time from exposure until disease onset; time from onset until recurrence; time from disease onset until disease progression; time from onset until death	survival analysis
ordered levels[2]	pain level (0 to 10 scale); approval rating (0 to 5 stars)	ordinal regression

[1] https://r4ds.hadley.nz/quarto
[2] Time permitting.

5 The scientific method and models

As scientists, we attempt to answer factual questions about the world and universe in the past, present, and/or future. In this course, we follow the steps of the scientific method[3], with a statistical model standing in for the hypothesis:

- conceive a question
- hypothesize a specific version of that model
- propose a study design that would provide data relevant to the model
- analyze the data
- estimate model parameters
- "inference":
  - quantify uncertainty
  - draw conclusions

Regression models are the family of models we use for these steps.

6 Motivations for regression models

Exercise 6.1. Why do we need regression models?

Solution

Solution 6.1.

- when there's not enough data to analyze every subgroup of interest individually
- especially when subgroups are defined using continuous predictors

---

6.1 Uses of regression models

Vittinghoff et al. (2012, sec. 1.3) identifies three broad motivations for using multipredictor regression models:

> Multipredictor regression can be a powerful tool for addressing three important practical questions. … [These] include prediction, isolating the effect of a single predictor, and understanding multiple predictors.
>
> 1. **Prediction**: "Multipredictor regression is a powerful and general tool for using multiple measured predictors to make useful predictions for future observations."
> 2. **Isolating the Effect of a Single Predictor**: "In settings where multiple, related predictors contribute to study outcomes, it will be important to consider multiple predictors even when a single predictor is of interest." (e.g., to minimize confounding and support causal interpretation)
> 3. **Understanding Multiple Predictors**: "Multipredictor regression can also be used when our aim is to identify multiple independent predictors of a study outcome — independent in the sense that they appear to have an effect over and above other measured variables." (including mediation and interaction)

Kleinbaum et al. (2014, sec. 4.1) provides a more granular list of eight overlapping applications of regression analysis:

> In practice, a regression analysis is appropriate for several possibly overlapping situations, including the following:

[3] https://en.wikipedia.org/wiki/Scientific_method

1. **Characterize the association**: "You want to characterize the relationship between the dependent and independent variables by determining the extent, direction, and strength of the association."
2. **Prediction**: "You seek a quantitative formula or equation to describe (e.g., predict) the dependent variable Y as a function of the independent variables $X_1, X_2, \ldots, X_k$."
3. **Controlled description**: "You want to describe quantitatively or qualitatively the relationship between $X_1, X_2, \ldots, X_k$ and Y but control for the effects of still other variables $X_{k+1}, X_{k+2}, \ldots, X_{k+p}$, which you believe have an important relationship with the dependent variable."
4. **Variable selection**: "You want to determine which of several independent variables are important and which are not for describing or predicting a dependent variable. You may want to control for other variables. You may also want to rank independent variables in their order of importance."
5. **Model selection**: "You want to determine the best mathematical model for describing the relationship between a dependent variable and one or more independent variables."
6. **Comparing regression relationships**: "You want to compare several derived regression relationships." (e.g., whether a relationship between two variables differs across subgroups)
7. **Interaction**: "You want to assess the interactive effects of two or more independent variables with regard to a dependent variable."
8. **Adjusted coefficient estimation**: "You want to obtain a valid and precise estimate of one or more regression coefficients from a larger set of regression coefficients in a given model." (i.e., estimating the effect of one variable after adjusting for others)

6.1.1 Relating the two lists

The two lists use different levels of granularity to describe the same landscape of regression uses. Vittinghoff et al. (2012) provides three broad categories, while Kleinbaum et al. (2014) identifies eight more specific applications.

Table 2: Comparing the categorizations of regression model uses in Vittinghoff et al. (2012, sec. 1.3) and Kleinbaum et al. (2014, sec. 4.1)

Vittinghoff et al. (2012) category	Kleinbaum et al. (2014) application(s)
Prediction	Application 2 (prediction)
Isolating the Effect of a Single Predictor	Application 8 (adjusted coefficient estimation)
Understanding Multiple Predictors	Application 1 (characterize association), Application 3 (controlled description), Application 4 (variable selection), Application 5 (model selection), Application 6 (comparing regression relationships), Application 7 (interaction)

Applications 6 and 7 are related but distinct: Application 6 asks whether a derived regression relationship (e.g., a coefficient or the overall model) differs across pre-defined groups, typically by comparing models fit separately for each group. Application 7 asks whether two predictors interact within a single model — that is, whether the effect of one predictor on the outcome depends on the value of another predictor. In that sense, Application 6 can be viewed as a special case of Application 7 where the grouping variable is the effect modifier.

The key conceptual distinction made by Vittinghoff et al. (2012) — but not explicitly highlighted by Kleinbaum et al. (2014) — is between **prediction** (forecasting future outcomes) and **causal**

inference (estimating the effect of a specific predictor). This distinction has important implications for model building strategy: prediction models can include any variables that improve predictive accuracy, while causal inference requires careful consideration of confounding, mediation, and the causal structure of the data.

6.1.2 Prediction, Inference, and Causal Inference

The goal of **prediction** is to forecast an outcome Y for new observations, given observed predictor variables $X_1, \ldots, X_p$. In a prediction problem, the model is often treated as a "black box": we care about predictive accuracy, not about the exact form or interpretation of the relationship between predictors and outcome (James et al. 2021, chap. 2).

Exm

Example 6.1. For example, a clinical prediction model might identify which patients are at highest risk of hospital readmission, using demographics, diagnoses, and laboratory values, without requiring that each predictor be causally related to readmission.

At its core, prediction aspires to minimize uncertainty: the goal is that, by including enough informative predictors and combining them flexibly, we can reduce as much as possible the residual randomness in our forecasts.

Statistical inference also studies the relationship between Y and $X_1, \ldots, X_p$, but the goal shifts from accuracy to understanding: which predictors are associated with the response, in what direction, and by how much (James et al. 2021, chap. 2)?

Exm

Example 6.2. For example, a cohort study might estimate the association between smoking and lung cancer incidence, along with a confidence interval quantifying uncertainty about the magnitude of that association.

Unlike prediction, statistical inference treats the uncertainty in outcomes as a meaningful quantity to characterize, not merely residual noise to be minimized.

Causal inference asks a distinct third question. A predictive association between a predictor X and an outcome Y does not imply that intervening to change X will change Y. Causal inference addresses this gap: it asks what would happen to Y if we were to actively *set* X to a particular value, rather than passively observe it (Vittinghoff et al. 2012, chap. 12).

Exm

Example 6.3. For example, even if smoking is strongly associated with lung cancer in observational data, causal inference asks whether reducing smoking rates would *cause* a reduction in lung cancer incidence — a question that requires additional assumptions about the data-generating process.

One useful framing of causal inference is as a prediction problem in **counterfactual** worlds: predicting what outcomes would have occurred under treatment conditions that were not actually administered. Under the **potential outcomes framework** (Neyman 1990; Rubin 1974, 688–701; Vittinghoff et al. 2012, chap. 12), individual causal effects are treated as deterministic in principle — each person has a fixed potential outcome under each treatment condition — and uncertainty arises from the fact that we can observe only one potential outcome per person, and from heterogeneity in effects across individuals.

Accurate causal inference typically requires accounting for **confounding**: in observational data, people who receive a treatment are often systematically different from those who do not in ways that also affect the outcome. These differences can make predictive associations misleading for causal questions.

A useful informal distinction: *prediction* asks which individuals are at highest risk, while *causal inference* asks what is the effect of a particular intervention, and *statistical inference* asks what the association is without necessarily specifying an intervention.

For an accessible introduction to the distinction between prediction and statistical inference, see James et al. (2021, chap. 2). For coverage of causal inference methods in an epidemiological setting, see Vittinghoff et al. (2012, chap. 12) and Hernán and Robins (2020, pt. I). For an in-depth treatment, see the causal inference chapter[4].

6.2 Example: Adelie penguins

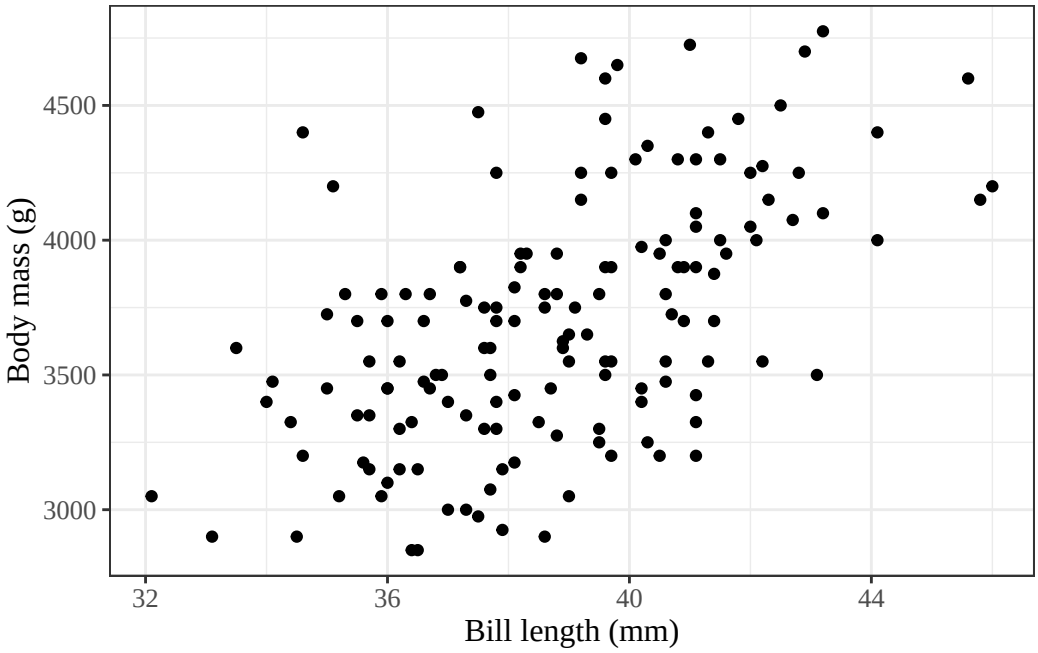


Figure 1: Palmer penguins

6.3 Linear regression

```
ggpenguins2 <-
  ggpenguins +
  stat_smooth(
    method = "lm",
    formula = y ~ x,
    geom = "smooth"
  )

ggpenguins2 |> print()
```

[4] causal-inference.qmd

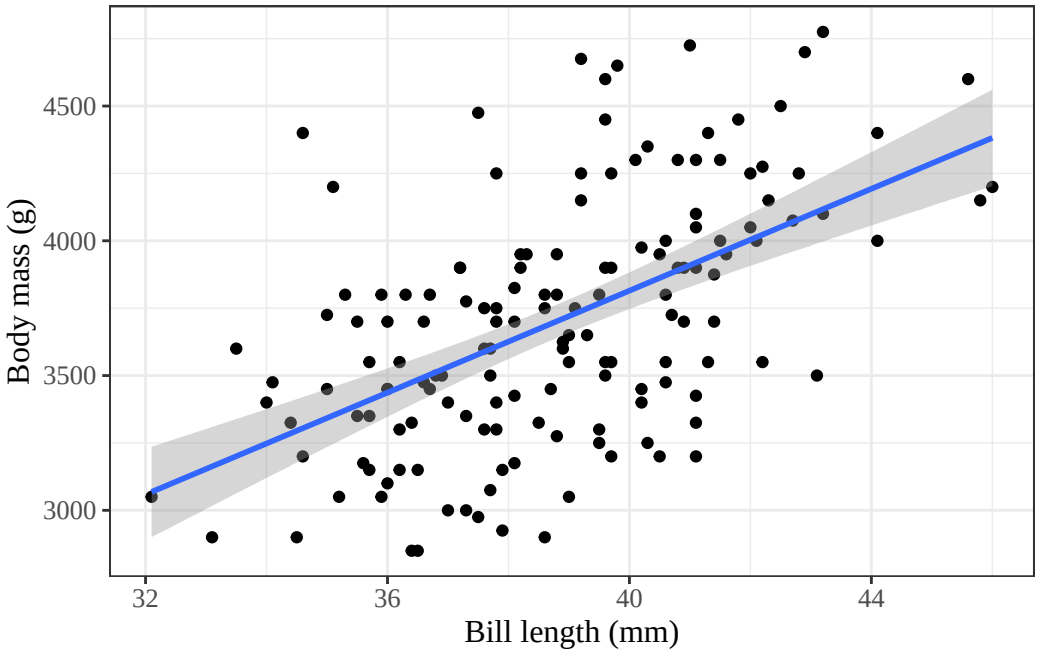


Figure 2: Palmer penguins with linear regression fit

6.4 Curved regression lines

```
ggpenguins2 <- ggpenguins +
  stat_smooth(
    method = "lm",
    formula = y ~ log(x),
    geom = "smooth"
  ) +
  xlab("Bill length (mm)") +
  ylab("Body mass (g)")
ggpenguins2
```

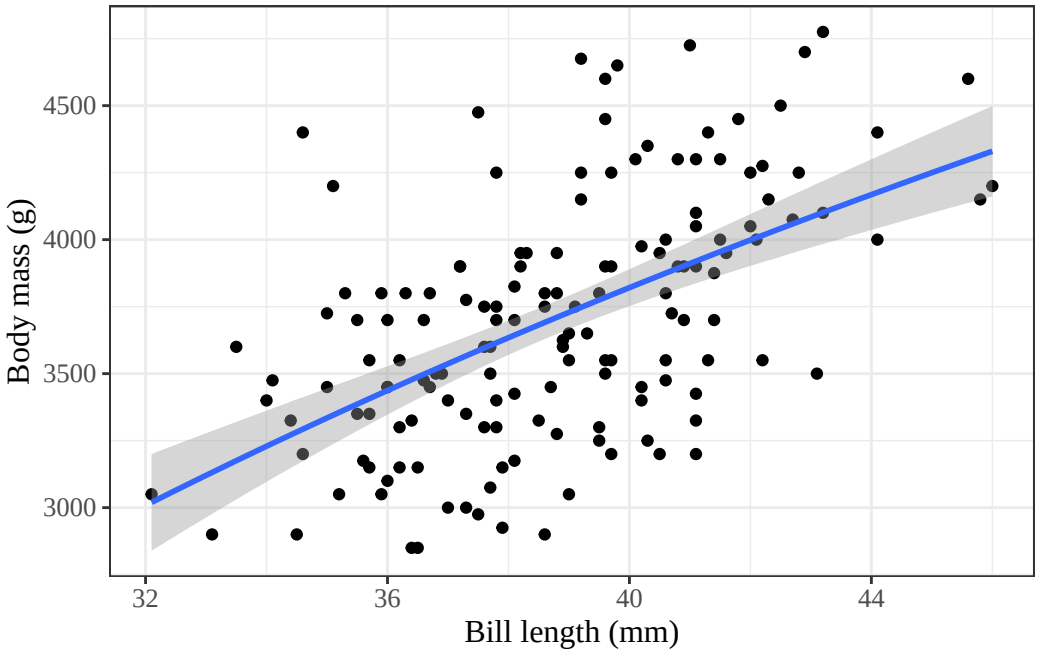


Figure 3: Palmer penguins - curved regression lines

6.5 Multiple regression

```
ggpenguins <-
  palmerpenguins::penguins |>
  ggplot(
    aes(
      x = bill_length_mm,
      y = body_mass_g,
      color = species
    )
  ) +
  geom_point() +
  stat_smooth(
    method = "lm",
    formula = y ~ x,
    geom = "smooth"
  ) +
  xlab("Bill length (mm)") +
  ylab("Body mass (g)")
ggpenguins |> print()
```

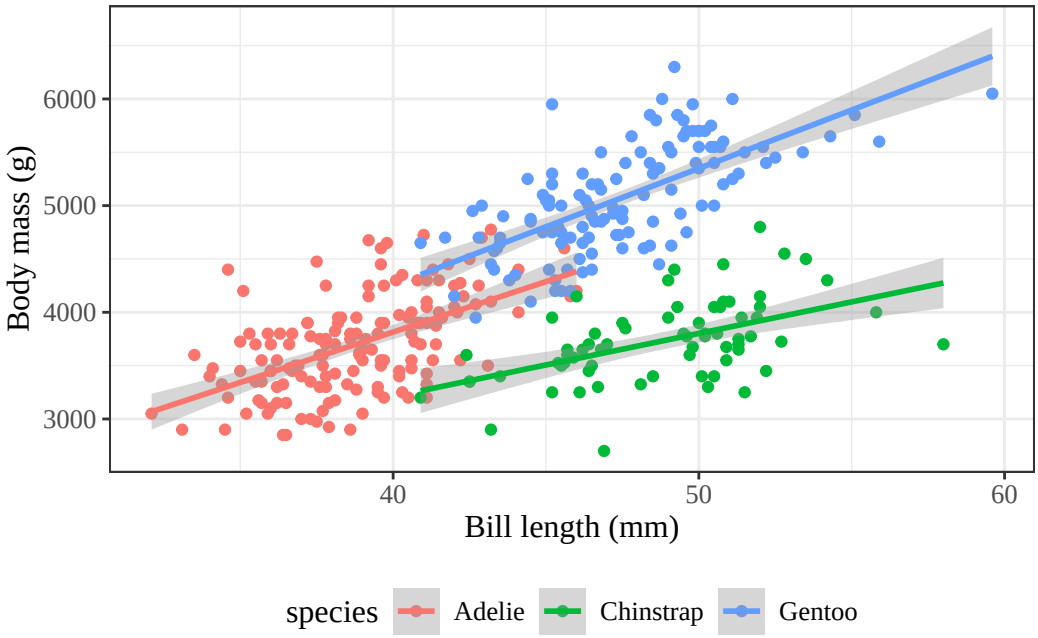


Figure 4: Palmer penguins - multiple groups

6.6 Modeling non-Gaussian outcomes

```
library(glmx)
data(BeetleMortality)
beetles <- BeetleMortality |>
  mutate(
    pct = died / n,
    survived = n - died
  )

plot1 <-
  beetles |>
  ggplot(aes(x = dose, y = pct)) +
  geom_point(aes(size = n)) +
  xlab("Dose (log mg/L)") +
  ylab("Mortality rate (%)") +
  scale_y_continuous(labels = scales::percent) +
  # xlab(bquote(log[10]), bquote(CS[2])) +
  scale_size(range = c(1, 2))

print(plot1)
```

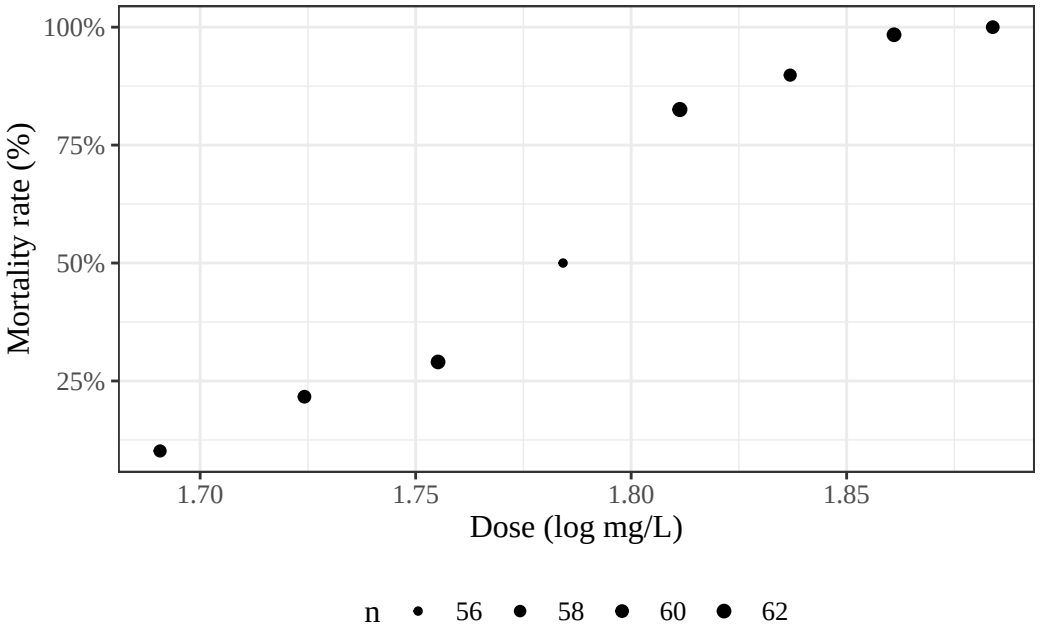


Figure 5: Mortality rates of adult flour beetles after five hours' exposure to gaseous carbon disulphide (Bliss 1935)

6.7 Why don't we use linear regression?

```
beetles_long <-
  beetles |>
  reframe(
    .by = everything(),
    outcome = c(
      rep(1, times = died),
      rep(0, times = survived)
    )
  )

lm1 <-
  beetles_long |>
  lm(
    formula = outcome ~ dose,
    data = _
  )


range1 <- range(beetles$dose) + c(-.2, .2)

f_linear <- function(x) predict(lm1, newdata = data.frame(dose = x))

plot2 <-
  plot1 +
  geom_function(fun = f_linear, aes(col = "Straight line")) +
  labs(colour = "Model", size = "")
print(plot2)
```

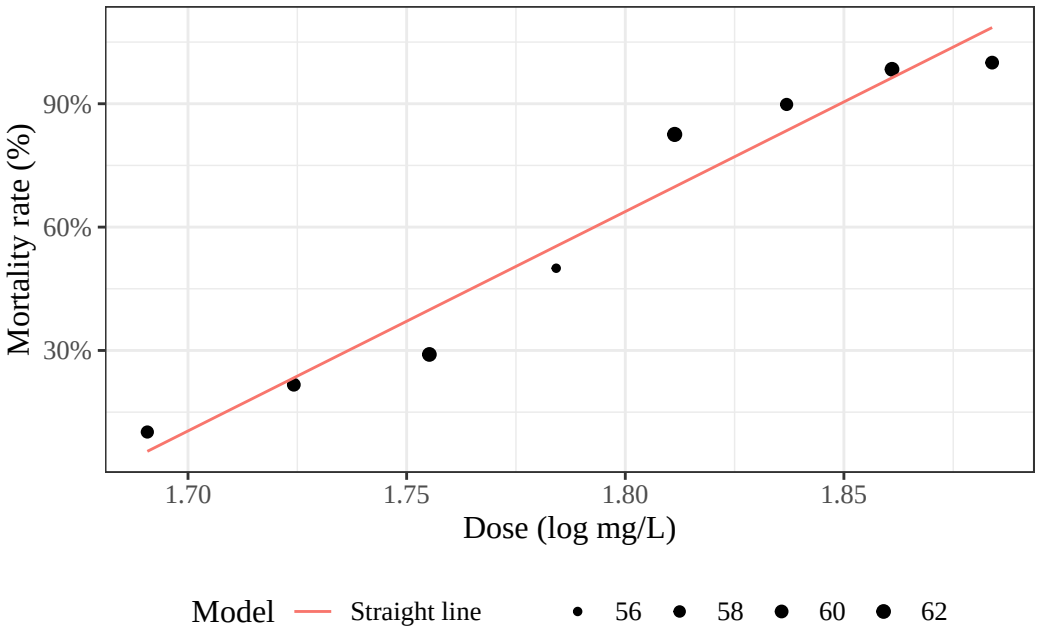


Figure 6: Mortality rates of adult flour beetles after five hours' exposure to gaseous carbon disulphide (Bliss 1935)

6.8 Zoom out

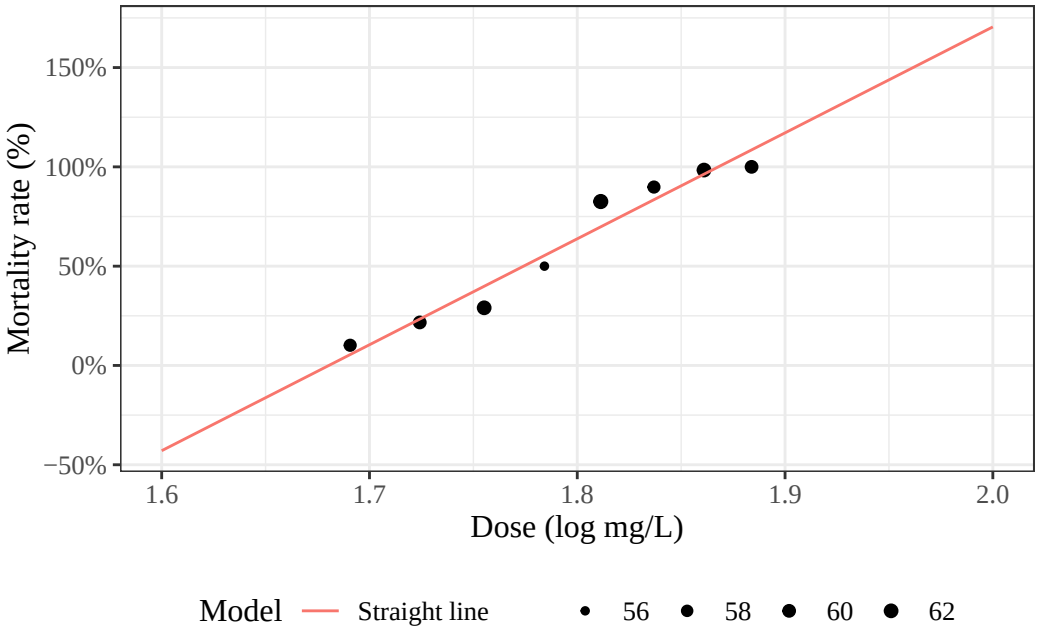


Figure 7: Mortality rates of adult flour beetles after five hours' exposure to gaseous carbon disulphide (Bliss 1935)

6.9 log transformation of dose?

```
lm2 <-
  beetles_long |>
  lm(formula = outcome ~ log(dose), data = _)

f_linearlog <- function(x) predict(lm2, newdata = data.frame(dose = x))

plot3 <- plot2 +
  expand_limits(x = c(1.6, 2)) +
  geom_function(fun = f_linearlog, aes(col = "Log-transform dose"))

print(plot3 + expand_limits(x = c(1.6, 2)))
```

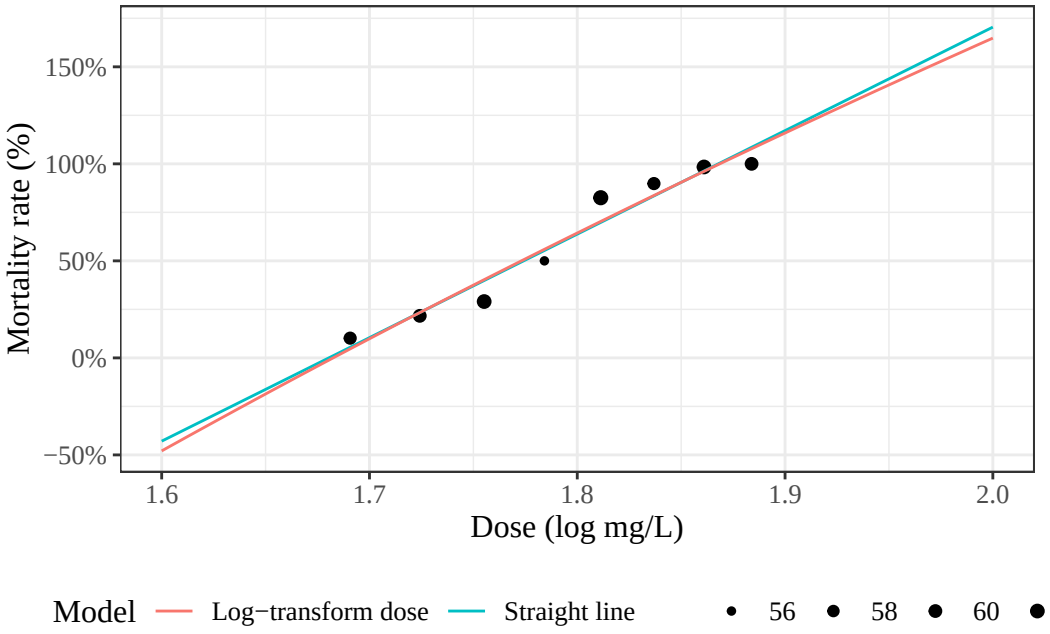


Figure 8: Mortality rates of adult flour beetles after five hours' exposure to gaseous carbon disulphide (Bliss 1935)

6.10 Logistic regression

```
glm1 <- beetles |>
  glm(formula = cbind(died, survived) ~ dose, family = "binomial")

f <- function(x) {
  glm1 |>
    predict(newdata = data.frame(dose = x), type = "response")
}

plot4 <- plot3 + geom_function(fun = f, aes(col = "Logistic regression"))
print(plot4)
```

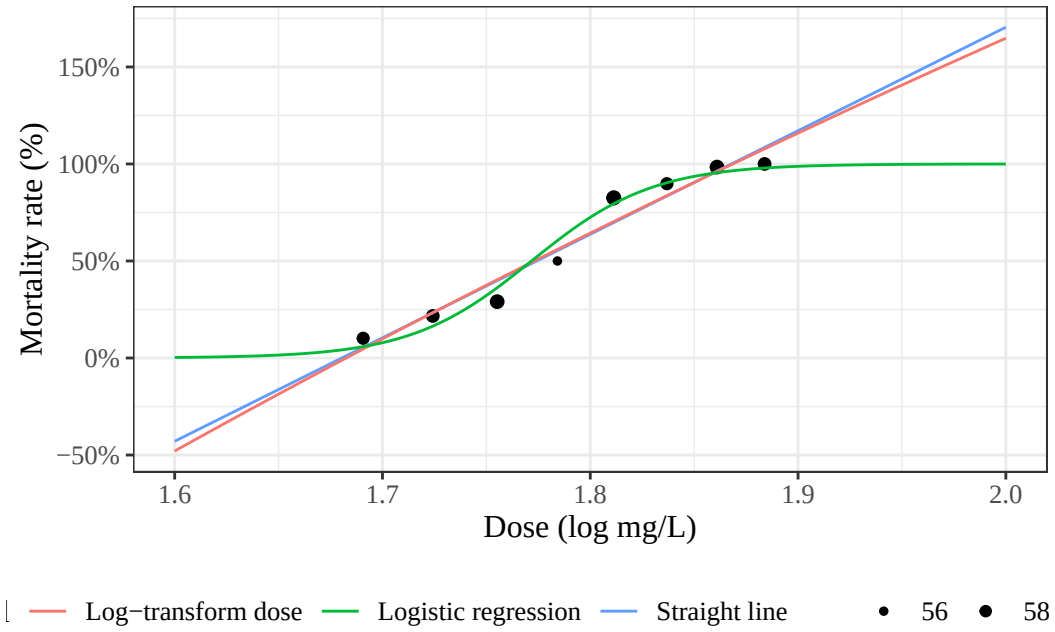


Figure 9: Mortality rates of adult flour beetles after five hours' exposure to gaseous carbon disulphide (Bliss 1935)

7 Structure of regression models

Exercise 7.1. What is a regression model?

Definition 7.1 (Regression model)**.** Regression models are conditional probability distribution models:

$$\mathrm{P}(Y|\tilde{X})$$

Exercise 7.2. What are some of the names used for the variables in a regression model $\mathrm{P}(Y|\tilde{X})$?

Definition 7.2 (Outcome)**.** The outcome variable in a regression model is the variable whose distribution is being described; in other words, the variable on the left-hand side of the "|" ("pipe") symbol.
The outcome variable is also called the **response variable**, **regressand**, **predicted variable**, **explained variable**, **experimental variable**, **output variable**, **dependent variable**, **endogenous variables**, **target**, or **label**.
and is typically denoted Y.

Definition 7.3 (Predictors)**.** The predictor variables in a regression model are the conditioning variables defining subpopulations among which the outcome distribution might vary.
Predictors are also called **regressors**, **covariates**, **independent variables**, **explanatory variables**, **risk factors**, **exposure variables**, **input variables**, **exogenous variables**, **candidate variables** (Dunn and Smyth (2018)), **carriers** (Dunn and Smyth (2018)), **manipulated variables**, or **features** and are typically denoted $\tilde{X}$. [5]

Table 3: Common pairings of terms for variables $\tilde{X}$ and Y in regression models $P(Y|\tilde{X})$ [7]

$\tilde{X}$	Y	usual context
input	output	
independent	dependent	
predictor	predicted or response	
explanatory	explained	
exogenous	endogenous	econometrics
manipulated	measured	randomized controlled experiments
exposure	outcome	epidemiology
feature	label or target	machine learning

Exercise 7.3. What is the general structure of a generalized linear model?

Solution

Solution 7.1. Generalized linear models have three components:

1. The **outcome distribution** family: $\mathrm{p}(Y|\mu(\tilde{x}))$
2. The **link function**: $g(\mu(\tilde{x})) = \eta(\tilde{x})$
3. The **linear component**: $\eta(\tilde{x}) = \tilde{x} \cdot \beta$

---

1. The **outcome distribution** family (a.k.a. the **random component** of the model)
- Gaussian (normal)
- Binomial
- Poisson
- Exponential
- Gamma
- Negative binomial

---

2. The **linear component** (a.k.a. the *linear predictor* or *linear functional form*) describing how the covariates combine to define subpopulations:

[5] The "~" ("tilde") symbol in the notation $\tilde{X}$ indicates that $\tilde{X}$ is a vector. See the notation page of the lab's math notes[6] for a table of notation used in these notes.
[7] adapted from https://en.wikipedia.org/wiki/Dependent_and_independent_variables#Synonyms

$$\eta(\tilde{x}) \stackrel{\text{def}}{=} \tilde{x}^\top \tilde{\beta} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ...$$

---

3. The **link function** relating the outcome distribution to the linear component, typically through the mean:

- identity: $\mu(y) = \eta(\tilde{x})$
- logit: $\log\left\{\frac{\mu(y)}{1-\mu(y)}\right\} = \eta(\tilde{x})$
- log: $\log\{\mu(y)\} = \eta(\tilde{x})$
- inverse: $(\mu(y))^{-1} = \eta(\tilde{x})$
- clog-log: $\log\{-\log\{1-\mu(y)\}\} = \eta(\tilde{x})$

Components 2 and 3 together are sometimes called the **systematic component** of the model (for example, in Dunn and Smyth (2018)).

8 Why link functions?

So far, we have seen models for Gaussian distributions that vary across subpopulations, and distributions for single populations. A GLM combines the two: it describes how a non-Gaussian distribution varies across subpopulations. The link function is the piece that makes this work.

Exercise 8.1. Suppose the outcome is a count or a binary variable. Why not model the mean $\mu(\tilde{x})$ as a linear function of the covariates, as in a Gaussian model?

Solution

Solution 8.1. A straight line for $\mu(\tilde{x})$ can leave the range of the outcome: it can predict a negative count, or a probability below 0 or above 1. We could try to repair this by transforming each covariate separately and adding interaction terms until the fitted curve stays in range, but that is hard to get right.

The link function offers a simpler route. We add all the covariate terms together first, to get the linear predictor $\eta(\tilde{x})$, and then transform once, with the inverse link g^{-1}. The link g relates the mean to the linear predictor: $g(\mu(\tilde{x})) = \eta(\tilde{x})$. Applying g^{-1} to both sides gives:

$$\mu(\tilde{x}) = g^{-1}(\eta(\tilde{x}))$$

Choosing the inverse link guarantees the model structure we want:

- with the log link, $\mu(\tilde{x}) = \exp\{\eta(\tilde{x})\} > 0$, so a mean count is never negative;
- with the logit link, $\mu(\tilde{x}) = \text{expit}(\eta(\tilde{x}))$ lies in $(0, 1)$, so a probability is never below 0 or above 1.

Remark 8.1 (Slopes under a non-identity link). With a non-identity link, interpreting a slope gets harder. A coefficient β_j is the change in $g(\mu)$ per unit change in x_j, not the change in μ. Even without any interaction terms, the change in μ per unit change in x_j depends on the values of the other covariates, because g^{-1} is not linear.

Exm

Example 8.1 (A logistic model with no interaction)**.** Suppose the mean μ of a binary outcome, which is its probability, satisfies $\text{logit}(\mu) = -2 + 1 \cdot x + 1.5 \cdot z$, with $z \in \{0, 1\}$. The coefficient of x is the same at both values of z, so the odds ratio for a one-unit increase in x is e^1 at both values of z. The code below computes the change in probability for the same increase in x, from $x = 0$ to $x = 1$.

```r
risk_diff <- function(z) {
  plogis(-2 + 1 * 1 + 1.5 * z) - plogis(-2 + 1 * 0 + 1.5 * z)
}
rd0 <- risk_diff(0)
rd1 <- risk_diff(1)
```

For $z = 0$, the probability rises by 0.15; for $z = 1$, it rises by 0.245. The odds ratio is the same in both subgroups, but the difference in probabilities is not.

9 Canonical links

The link functions listed in the structure of a GLM are not all on the same footing. For each outcome distribution, the form of its density singles out one link. To say which, we need the density's *natural parameter*.

Definition 9.1 (Exponential dispersion family, natural parameter, and canonical link)**.** A distribution belongs to the **exponential dispersion family** if its density (or probability mass function) can be written as

$$f(y; \theta, \phi) = \exp\left\{ \frac{y\,\theta - b(\theta)}{\phi} + c(y, \phi) \right\}$$

for some functions $b(\cdot)$ and $c(\cdot, \cdot)$. The parameter θ is the **natural parameter**, and $\phi > 0$ is the **dispersion parameter**. Then

$$\begin{aligned} \mathrm{E}[Y] &= b'(\theta) \stackrel{\text{def}}{=} \mu \\ \mathrm{Var}(Y) &= \phi\, b''(\theta) = \phi\, V(\mu), \end{aligned}$$

where $V(\mu) \stackrel{\text{def}}{=} b''(\theta)$, written as a function of μ, is the **variance function**. The **canonical link** is the link function $g(\cdot)$ that sets the linear predictor equal to the natural parameter: $g(\mu) \stackrel{\text{def}}{=} \theta$ (Dobson and Barnett 2018, chap. 3; Dunn and Smyth 2018, chap. 5).

Exm

Example 9.1 (Natural parameters of a Poisson and a Bernoulli outcome)**.** For a Poisson outcome with mean μ,

$$f(y; \mu) = \frac{\mu^y e^{-\mu}}{y!} = \exp\{y \log\{\mu\} - \mu - \log\{y!\}\},$$

so $\theta \stackrel{\text{def}}{=} \log\{\mu\}$, $b(\theta) \stackrel{\text{def}}{=} e^\theta$, and $\phi \stackrel{\text{def}}{=} 1$. For a Bernoulli outcome with success probability π,

$$f(y; \pi) = \pi^y (1 - \pi)^{1-y} = \exp\left\{ y \log\left\{ \frac{\pi}{1 - \pi} \right\} + \log\{1 - \pi\} \right\},$$

so $\theta \stackrel{\text{def}}{=} \mathrm{logit}(\pi)$, $b(\theta) \stackrel{\text{def}}{=} \log\{1 + e^\theta\}$, and $\phi \stackrel{\text{def}}{=} 1$.
The code below checks both forms against R's built-in densities, at $\mu = 3$, $y = 2$ and at $\pi = 0.25$, $y = 1$, and computes $b'(\theta)$ and $b''(\theta)$ for the Bernoulli case.

```r
mu <- 3
theta_pois <- log(mu)
y_pois <- 2
pois_exp_form <- exp(y_pois * theta_pois - exp(theta_pois) - lgamma(y_pois + 1))
pois_check <- isTRUE(all.equal(pois_exp_form, dpois(y_pois, mu)))

pi_bern <- 0.25
theta_bern <- qlogis(pi_bern)
y_bern <- 1
bern_exp_form <- exp(y_bern * theta_bern - log1p(exp(theta_bern)))
bern_check <- isTRUE(all.equal(bern_exp_form, dbinom(y_bern, 1, pi_bern)))

b1_bern <- plogis(theta_bern)
b2_bern <- b1_bern * (1 - b1_bern)
```

- For the Poisson distribution with $\mu = 3$, the natural parameter is $\theta = \log(3) = 1.0986$, and the exponential-family form gives $f(2) = 0.224$, which matches `dpois(2, 3)`: TRUE. Here $b'(\theta) = b''(\theta) = e^{\theta} = 3$, so $V(\mu) = \mu$.
- For the Bernoulli distribution with $\pi = 0.25$, the natural parameter is $\theta = \text{logit}(0.25) =$ -1.0986, and the exponential-family form gives $f(1) = 0.25$, which matches `dbinom(1, 1, 0.25)`: TRUE. Here $b'(\theta) = \text{expit}(\theta) = 0.25$, which is π, and $b''(\theta) = \pi(1-\pi) = 0.1875$, so $V(\pi) = \pi(1-\pi)$.

Consequently, the canonical link for the Poisson distribution is the log link, and the canonical link for the Bernoulli distribution is the logit link.

10 The model fitting process

The rest of this chapter is adapted from Dobson and Barnett (2018, chap. 2).

Dobson and Barnett (2018, sec. 2.1) describes the model fitting process as follows:

> The model fitting process described in this book involves four steps:
>
> 1. Model specification—a model is specified in two parts: an equation linking the response and explanatory variables and the probability distribution of the response variable.
> 2. Estimation of the parameters of the model.
> 3. Checking the adequacy of the model—how well it fits or summarizes the data.
> 4. Inference—for classical or frequentist inference this involves calculating confidence intervals, testing hypotheses about the parameters in the model and interpreting the results.

Note that Dobson and Barnett (2018) omits exploratory data analysis (EDA) from this initial list, but describes it later as an important preliminary step (Dobson and Barnett (2018, sec. 2.3.1)). We will treat EDA as a "step 0" that precedes these four steps.

10.1 Step 0: Exploratory data analysis

Dobson and Barnett (2018, sec. 2.3.1):

> Any analysis of data should begin with a consideration of each variable separately, both to check on data quality (for example, are the values plausible?) and to help with model formulation.
>
> 1. What is the scale of measurement? Is it continuous or categorical? If it is categorical, how many categories does it have and are they nominal or ordinal?
> 2. What is the shape of the distribution? This shape can be examined using frequency tables, dot plots, histograms and other graphical methods.

3. How is it associated with other variables? Cross tabulations for categorical variables, scatter plots for continuous variables, side-by-side boxplots for continuous scale measurements grouped according to the factor levels of a categorical variable, and other such summaries can help to identify patterns of association. For example, do the points on a scatter plot suggest linear or non-linear associations? Do the group means increase or decrease consistently with an ordinal variable defining the groups?

Exm

Example 10.1 (Exploring the chronic medical conditions data)**.** Dobson and Barnett (2018, sec. 2.2.1, Table 2.1) reports the number of chronic medical conditions for 26 women living in a town and 23 women living in the country, all attending general practitioners. We will carry this dataset through all four steps of the model fitting process.

Symbol	Meaning	Column in `dobson::chronic`
Y_i	number of chronic medical conditions reported by woman i	`number` (copied to `conditions`)
x_i	indicator that woman i lives in the country (1) rather than the town (0)	derived from `place` (stored as `country`)

```
data(chronic, package = "dobson")
chronic$conditions <- chronic$number
chronic$country <- as.numeric(chronic$place == "country")
chronic$residence <- chronic$place
```

Following the three questions of Dobson and Barnett (2018, sec. 2.3.1):

1. The outcome Y is a count, so its scale of measurement is discrete and non-negative; the predictor x is a binary (nominal) indicator.
2. Table 4 shows the frequency distribution of Y within each group. Both distributions are right-skewed, with most women reporting zero, one, or two conditions.
3. The town women report more conditions on average (sample mean 1.423) than the country women (sample mean 0.913), so residence might be associated with the outcome.

Table 4: Frequency distribution of the number of chronic medical conditions, by residence (Dobson and Barnett 2018, sec. 2.2.1, Table 2.1)

```
table(
  residence = chronic$residence,
  conditions = chronic$conditions
)
#>           conditions
#> residence  0  1  2  3  4
#>   town     6 10  4  5  1
#>   country  9  8  5  1  0
```

10.2 Step 1: Model specification

Dobson and Barnett (2018, sec. 2.3.2) describes model specification as choosing two parts: an equation linking the expected outcome to the explanatory variables, and a probability distribution for the outcome. For a generalized linear model, those two parts are the three components in the solution to Exercise 7.3: an outcome distribution, a linear predictor $\eta \stackrel{\text{def}}{=} \tilde{x} \cdot \tilde{\beta}$, and a link function $g(\cdot)$ with $g(\mu) = \eta$.

Exm

Example 10.2 (Specifying a model for the chronic medical conditions data)**.** The outcome in Example 10.1 is a count with no fixed upper limit, so a Poisson distribution is a natural first choice. Since a Poisson mean must be positive, we pair the Poisson distribution with the log link (which is also its canonical link; see Section 9):

$$\begin{aligned} Y_i \mid x_i &\sim_{\perp\!\!\!\perp} \text{Pois}(\mu_i), \quad i = 1, \ldots, 49 \\ \log\{\mu_i\} &= \eta_i \\ \eta_i &\stackrel{\text{def}}{=} \beta_0 + \beta_1 x_i \end{aligned}$$

Under this model, $\mu_{\text{town}} \stackrel{\text{def}}{=} \exp\{\beta_0\}$ is the mean number of conditions for town women, and $\mu_{\text{country}} \stackrel{\text{def}}{=} \exp\{\beta_0 + \beta_1\}$ is the mean for country women, so the ratio of the two group means is

$$\begin{aligned} \frac{\mu_{\text{country}}}{\mu_{\text{town}}} &= \frac{\exp\{\beta_0 + \beta_1\}}{\exp\{\beta_0\}} && \text{(definitions of the two group means)} \\ &= \frac{\exp\{\beta_0\}\exp\{\beta_1\}}{\exp\{\beta_0\}} && (e^{a+b} = e^a e^b) \\ &= \exp\{\beta_1\} && \text{(cancel } \exp\{\beta_0\}) \end{aligned}$$

10.3 Step 2: Parameter estimation

Dobson and Barnett (2018, sec. 2.3.3) describes the next step as estimating the model's parameters, for example by maximum likelihood or by least squares. The model structure from Step 1 does not dictate the estimation method; in this chapter we use maximum likelihood, which chooses the parameter values that make the observed data most probable.

Exm

Example 10.3 (Maximum likelihood estimates for the chronic medical conditions model)**.** Write $n_{\text{town}} \stackrel{\text{def}}{=} 26$ and $n_{\text{country}} \stackrel{\text{def}}{=} 23$ for the group sizes, and $\bar{y}_{\text{town}}$ and $\bar{y}_{\text{country}}$ for the group sample means. For the model in Example 10.2, the log-likelihood is

$$\begin{aligned} \ell(\beta_0, \beta_1) &\stackrel{\text{def}}{=} \sum_{i=1}^{49} \log\left\{\frac{\mu_i^{y_i} e^{-\mu_i}}{y_i!}\right\} && \text{(independent Poisson outcomes)} \\ &= \sum_{i=1}^{49} [y_i \log\{\mu_i\} - \mu_i - \log\{y_i!\}] && \text{(log of a product and a quotient)} \\ &= \sum_{i=1}^{49} [y_i \eta_i - e^{\eta_i} - \log\{y_i!\}] && \text{(log link: } \log\{\mu_i\} = \eta_i) \\ &= \sum_{i=1}^{49} [y_i(\beta_0 + \beta_1 x_i) - e^{\beta_0 + \beta_1 x_i} - \log\{y_i!\}] && \text{(definition of } \eta_i) \end{aligned}$$

Differentiating term by term gives the two components of the score function:

$$
\begin{aligned}
\frac{\partial \ell}{\partial \beta_0} &= \sum_{i=1}^{49} [y_i - e^{\beta_0 + \beta_1 x_i}] && (\partial \eta_i / \partial \beta_0 = 1) \\
&= \sum_{i=1}^{49} (y_i - \mu_i) && (\mu_i = e^{\eta_i}) \\
\frac{\partial \ell}{\partial \beta_1} &= \sum_{i=1}^{49} x_i [y_i - e^{\beta_0 + \beta_1 x_i}] && (\partial \eta_i / \partial \beta_1 = x_i) \\
&= \sum_{i=1}^{49} x_i (y_i - \mu_i) && (\mu_i = e^{\eta_i})
\end{aligned}
$$

Since $x_i = 1$ for country women and $x_i = 0$ for town women, setting $\partial \ell / \partial \beta_1 = 0$ gives

$$
\begin{aligned}
0 &= \sum_{i:\, x_i = 1} (y_i - e^{\beta_0 + \beta_1}) && \text{(drop the terms with } x_i = 0) \\
&= n_{\text{country}} \bar{y}_{\text{country}} - n_{\text{country}} e^{\beta_0 + \beta_1} && \text{(sum of } n_{\text{country}} \text{ terms)} \\
e^{\beta_0 + \beta_1} &= \bar{y}_{\text{country}} && \text{(rearrange and divide by } n_{\text{country}})
\end{aligned}
$$

Subtracting the second score equation from the first leaves only the town women, since $1 - x_i = 1$ exactly when woman i lives in the town:

$$
\begin{aligned}
0 &= \sum_{i=1}^{49} (1 - x_i)(y_i - \mu_i) && \text{(difference of the two score equations)} \\
&= \sum_{i:\, x_i = 0} (y_i - e^{\beta_0}) && \text{(drop the terms with } x_i = 1) \\
&= n_{\text{town}} \bar{y}_{\text{town}} - n_{\text{town}} e^{\beta_0} && \text{(sum of } n_{\text{town}} \text{ terms)} \\
e^{\beta_0} &= \bar{y}_{\text{town}} && \text{(rearrange and divide by } n_{\text{town}})
\end{aligned}
$$

Taking logs of both solutions gives the maximum likelihood estimates:

$$
\begin{aligned}
\hat{\beta}_0 &= \log\{\bar{y}_{\text{town}}\} \\
\hat{\beta}_1 &= \log\{\bar{y}_{\text{country}}\} - \hat{\beta}_0 && \text{(log of } e^{\beta_0 + \beta_1} = \bar{y}_{\text{country}}) \\
&= \log\{\bar{y}_{\text{country}}\} - \log\{\bar{y}_{\text{town}}\} && \text{(substitute } \hat{\beta}_0) \\
&= \log\left\{ \frac{\bar{y}_{\text{country}}}{\bar{y}_{\text{town}}} \right\} && \text{(difference of logs)}
\end{aligned}
$$

```
ybar_town <- mean(chronic$conditions[chronic$country == 0])
ybar_country <- mean(chronic$conditions[chronic$country == 1])
b0_hat <- log(ybar_town)
b1_hat <- log(ybar_country / ybar_town)

chronic_glm <- glm(
  conditions ~ country,
  family = poisson(link = "log"),
  data = chronic
)
coef(chronic_glm)
#> (Intercept)     country
#>    0.352821   -0.443793
```

With $\bar{y}_{\text{town}} = 37/26 \approx 1.4231$ and $\bar{y}_{\text{country}} = 21/23 \approx 0.9130$, the closed-form estimates are $\hat{\beta}_0 \approx 0.3528$ and $\hat{\beta}_1 \approx$ -0.4438, matching the coefficients that `glm()` finds numerically. The fitted means are the group sample means: $\hat{\mu}_i = \bar{y}_{\text{town}}$ for town women and $\hat{\mu}_i = \bar{y}_{\text{country}}$ for country women.

Exm

Example 10.4 (Checking that the chronic medical conditions estimates are a maximum)**.** To confirm that the estimates in Example 10.3 maximize the log-likelihood, differentiate the score components once more, using $\partial\mu_i/\partial\beta_0 = \mu_i$ and $\partial\mu_i/\partial\beta_1 = x_i\mu_i$ (chain rule applied to $\mu_i = e^{\beta_0+\beta_1 x_i}$):

$$\frac{\partial^2 \ell}{\partial \beta_0^2} = -\sum_{i=1}^{49} \mu_i$$
$$\frac{\partial^2 \ell}{\partial \beta_0\, \partial \beta_1} = -\sum_{i=1}^{49} x_i \mu_i$$
$$\frac{\partial^2 \ell}{\partial \beta_1^2} = -\sum_{i=1}^{49} x_i^2 \mu_i$$

Writing $\tilde{x}_i \overset{\text{def}}{=} (1, x_i)^\top$ and $\tilde{\beta} \overset{\text{def}}{=} (\beta_0, \beta_1)^\top$, these three second derivatives are the entries of the Hessian matrix

$$\underbrace{\ell''\big(\tilde{\beta}\big)}_{2\times 2} = -\sum_{i=1}^{49} \mu_i \underbrace{\tilde{x}_i}_{2\times 1} \underbrace{\tilde{x}_i^\top}_{1\times 2}, \quad \text{since} \quad \underbrace{\tilde{x}_i}_{2\times 1} \underbrace{\tilde{x}_i^\top}_{1\times 2} = \underbrace{\begin{pmatrix} 1 & x_i \\ x_i & x_i^2 \end{pmatrix}}_{2\times 2}.$$

For any nonzero vector $\underbrace{\tilde{a}}_{2\times 1}$, $\tilde{a}^\top \ell''\big(\tilde{\beta}\big)\tilde{a} = -\sum_{i=1}^{49} \mu_i (\tilde{a}^\top \tilde{x}_i)^2 < 0$, because every $\mu_i > 0$ and both groups contain women (so $\tilde{a}^\top \tilde{x}_i \neq 0$ for some i). The log-likelihood is therefore strictly concave, and the stationary point we found is its unique maximum.

10.4 Step 3: Checking model adequacy

Dobson and Barnett (2018, sec. 2.3.4) describes checking a fitted model's adequacy by examining residuals: summaries of how far each observation falls from its fitted value. In a generalized linear model the outcome variance usually changes with the mean, so raw differences $y_i - \hat{\mu}_i$ are not on a common scale; dividing by an estimated standard deviation puts them on one.

Definition 10.1 (Pearson residual and Pearson chi-squared statistic)**.** For a generalized linear model with variance function $V(\mu)$ (Definition 9.1) and dispersion parameter $\phi = 1$, the **Pearson residual** for observation i is

$$r_i \overset{\text{def}}{=} \frac{y_i - \hat{\mu}_i}{\sqrt{V(\hat{\mu}_i)}},$$

and the **Pearson chi-squared statistic** is $X^2 \overset{\text{def}}{=} \sum_{i=1}^{n} r_i^2$ (Dunn and Smyth 2018, chap. 7 and 8). For a Poisson outcome, $V(\mu) = \mu$.

If the model is adequate and the fitted means are not too small, X^2 is approximately chi-squared distributed with $n - p$ degrees of freedom, where p is the number of regression coefficients, so $X^2/(n-p)$ should be close to 1; a ratio well above 1 suggests overdispersion (Dunn and Smyth 2018, chap. 7). The count regression chapter[8] returns to overdispersion.

Exm

Example 10.5 (Pearson residuals for the chronic medical conditions model)**.** The town woman who reports 4 conditions has fitted mean $\hat{\mu}_i = \bar{y}_{\text{town}} = 37/26$ (Example 10.3), so her Pearson residual is

[8] count-regression.qmd

$$
\begin{aligned}
r_i &= \frac{4 - 37/26}{\sqrt{37/26}} && \text{(Pearson residual with } V(\mu) = \mu\text{)} \\
&\approx \frac{4 - 1.4231}{1.1929} && \text{(evaluate } 37/26 \text{ and its square root)} \\
&\approx \frac{2.5769}{1.1929} && \text{(subtract)} \\
&\approx 2.160 && \text{(divide)}
\end{aligned}
$$

```
r_pearson <- residuals(chronic_glm, type = "pearson")
x2 <- sum(r_pearson^2)
df_resid <- df.residual(chronic_glm)
```

Figure 10 plots all 49 Pearson residuals. The largest residual in absolute value is 2.184, and the Pearson chi-squared statistic is $X^2 \approx 43.66$ on $49 - 2 = 47$ residual degrees of freedom, so $X^2/47 \approx 0.929$. No residual is extreme, and the ratio is below 1, so the data show no sign of more variability than the Poisson model allows.

```
library(ggplot2)
chronic$pearson <- r_pearson
ggplot(chronic, aes(x = residence, y = pearson)) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  geom_jitter(width = 0.1, height = 0) +
  labs(x = "Residence", y = "Pearson residual") +
  theme_bw()
```

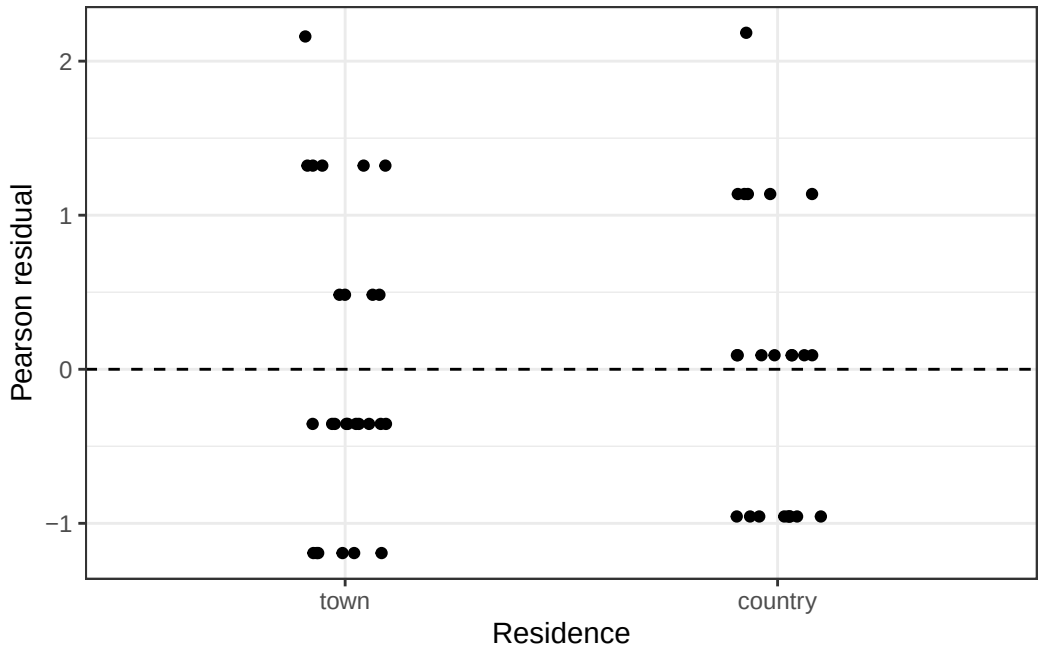


Figure 10: Pearson residuals from the Poisson model for the chronic medical conditions data, by residence

10.5 Step 4: Inference

The final step is inference: Dobson and Barnett (2018, sec. 2.3.5) lists confidence intervals, hypothesis tests about the parameters, and interpretation of the results. By the central limit theorem for MLEs[9], in large samples $\hat{\beta}$ is approximately Gaussian, with covariance matrix approximated by the inverse of the observed information $-\ell''\left(\hat{\beta}\right)$. A 95% Wald confidence interval for β_j is then

[9] https://morrison-lab.github.io/sds/intro-MLEs.html#thm-dist-mle

$\hat\beta_j \pm 1.96 \cdot \widehat{\mathrm{SE}}\big(\hat\beta_j\big)$, and the Wald statistic $z_j \overset{\text{def}}{=} \hat\beta_j / \widehat{\mathrm{SE}}\big(\hat\beta_j\big)$ tests $H_0 : \beta_j = 0$.

Exm

Example 10.6 (Comparing town and country women)**.** In Example 10.3, the fitted means sum to $\sum_{i:\,x_i=0} \hat\mu_i = n_{\text{town}} \bar y_{\text{town}} = 37$ for the town women and $\sum_{i:\,x_i=1} \hat\mu_i = n_{\text{country}} \bar y_{\text{country}} = 21$ for the country women. Substituting into the Hessian from Example 10.4, the observed information matrix is

$$
\begin{aligned}
\underbrace{-\ell''\big(\hat\beta\big)}_{2\times 2} &= \sum_{i=1}^{49} \hat\mu_i \begin{pmatrix} 1 & x_i \\ x_i & x_i^2 \end{pmatrix} && \text{(Hessian of the log-likelihood)} \\
&= \begin{pmatrix} \sum_i \hat\mu_i & \sum_{i:\,x_i=1} \hat\mu_i \\ \sum_{i:\,x_i=1} \hat\mu_i & \sum_{i:\,x_i=1} \hat\mu_i \end{pmatrix} && (x_i \in \{0,1\}, \text{ so } x_i^2 = x_i) \\
&= \underbrace{\begin{pmatrix} 58 & 21 \\ 21 & 21 \end{pmatrix}}_{2\times 2} && (37 + 21 = 58)
\end{aligned}
$$

Its determinant is $58 \cdot 21 - 21 \cdot 21 = 21 \cdot 37 = 777$, so its inverse is

$$
\underbrace{\big[-\ell''\big(\hat\beta\big)\big]^{-1}}_{2\times 2} = \frac{1}{777} \begin{pmatrix} 21 & -21 \\ -21 & 58 \end{pmatrix} \quad \text{(inverse of a } 2 \times 2 \text{ matrix)}
$$

The lower-right entry is the estimated variance of $\hat\beta_1$:

$$
\begin{aligned}
\widehat{\mathrm{Var}}\big(\hat\beta_1\big) &= \frac{58}{777} \\
&= \frac{37 + 21}{37 \cdot 21} && (58 = 37 + 21,\ 777 = 37 \cdot 21) \\
&= \frac{1}{21} + \frac{1}{37} && \text{(split the fraction)} \\
&\approx 0.0746
\end{aligned}
$$

so $\widehat{\mathrm{SE}}\big(\hat\beta_1\big) \approx \sqrt{0.0746} \approx 0.2732$.

```
se_b1 <- sqrt(1 / 21 + 1 / 37)
ci_b1 <- b1_hat + c(-1, 1) * qnorm(0.975) * se_b1
z_b1 <- b1_hat / se_b1
p_b1 <- 2 * pnorm(-abs(z_b1))
summary(chronic_glm)$coefficients
#>              Estimate Std. Error  z value  Pr(>|z|)
#> (Intercept)  0.352821   0.164399  2.14613 0.0318627
#> country     -0.443793   0.273214 -1.62434 0.1043026
```

- The 95% Wald confidence interval for β_1 is from -0.979 to 0.092.
- Exponentiating the estimate and the interval endpoints, the estimated ratio of the country mean to the town mean is $\exp\{\hat\beta_1\} \approx 0.642$, with 95% confidence interval from 0.376 to 1.096.
- The Wald statistic is $z \approx$ -1.624, with two-sided $p \approx 0.104$.

These hand calculations match the standard error, z value, and p-value that `summary()` reports for `country`. On average, country women in this sample reported 36% fewer chronic conditions than town women, but the confidence interval for the ratio includes 1, so these data are also consistent with no difference in the mean number of conditions.

11 Choosing a model

The choice of statistical model depends on several considerations, primary among which is the type of response variable.

- Measured continuous values such as protein level, age, or weight can usually be modeled using linear regression, possibly after applying a logarithmic transformation.
- Patient survival times, which may be right-censored, call for time-to-event methods such as Kaplan-Meier estimation or Cox proportional hazards regression.
- If the outcome is binary (0/1), we can use logistic regression models.
- If the outcome is a count, we can use Poisson regression.
- If count data exhibit greater variance than assumed under the Poisson model, we can use negative binomial regression or overdispersed Poisson models.
- Other outcome structures correspond to other families within the generalized linear model framework.

In generalized linear modeling, we specify a linear predictor $\eta \stackrel{\text{def}}{=} \tilde{x} \cdot \tilde{\beta}$ of the same form as in linear regression. In principle, a linear predictor can yield any real value, whether positive, negative, or zero.

We then select an appropriate outcome distribution matched to the data type (for example, Gaussian for unbounded continuous values, gamma for positive continuous quantities, binomial for binary outcomes, and Poisson for counts).

Finally, we specify a link function $g(\cdot)$ that maps the expected outcome $\mu \stackrel{\text{def}}{=} \mathrm{E}\left[Y \mid \tilde{X} = \tilde{x}\right]$ onto the real line $(-\infty, \infty)$ of the linear predictor. The inverse link function $g^{-1}(\cdot)$ transforms the linear predictor back to the scale of the expected outcome.

- Standard linear regression uses the identity link function $g(\mu) \stackrel{\text{def}}{=} \mu$ and assumes a Gaussian response distribution.
- When predicting an inherently positive outcome, a log link function $g(\mu) \stackrel{\text{def}}{=} \log\{\mu\}$ is often appropriate, since $\exp\{\eta\} > 0$ for all real η.
- An alternative to fitting a generalized linear model with a log link is to apply a log transformation directly to the outcome. While log-transforming the outcome works well for positive continuous measurements, it cannot be applied directly to binary outcomes or to count outcomes that can equal zero.

Table 5 lists the outcome families available in R's `glm()` function, together with the link functions each family accepts (the default link for each family is in bold), and Table 6 defines each link function.

Table 5: Families and link functions available in R's `glm()`

Family	Links
gaussian	**identity**, log, inverse
binomial	**logit**, probit, cauchit, log, cloglog
gamma	**inverse**, identity, log
inverse.gaussian	**1/mu^2**, inverse, identity, log
Poisson	**log**, identity, sqrt
quasi	**identity**, logit, probit, cloglog, inverse, log, 1/mu^2 and sqrt
quasibinomial	**logit**, probit, identity, cloglog, inverse, log, 1/mu^2 and sqrt
quasipoisson	**log**, identity, logit, probit, cloglog, inverse, 1/mu^2 and sqrt

Table 6: Link functions available in R's `glm()`, where $\eta \overset{\text{def}}{=} \tilde{x} \cdot \tilde{\beta} = g(\mu)$

Name	Domain	Range	Link Function	Inverse Link Function
identity	$(-\infty, \infty)$	$(-\infty, \infty)$	$\eta = \mu$	$\mu = \eta$
log	$(0, \infty)$	$(-\infty, \infty)$	$\eta = \log\{\mu\}$	$\mu = \exp\{\eta\}$
inverse	$(0, \infty)$	$(0, \infty)$	$\eta = 1/\mu$	$\mu = 1/\eta$
logit	$(0, 1)$	$(-\infty, \infty)$	$\eta = \log\{\mu/(1-\mu)\}$	$\mu = \exp\{\eta\}/(1+\exp\{\eta\})$
probit	$(0, 1)$	$(-\infty, \infty)$	$\eta = \Phi^{-1}(\mu)$	$\mu = \Phi(\eta)$
cloglog	$(0, 1)$	$(-\infty, \infty)$	$\eta = \log\{-\log\{1-\mu\}\}$	$\mu = 1 - \exp\{-\exp\{\eta\}\}$
1/mu^2	$(0, \infty)$	$(0, \infty)$	$\eta = 1/\mu^2$	$\mu = 1/\sqrt{\eta}$
sqrt	$(0, \infty)$	$(0, \infty)$	$\eta = \sqrt{\mu}$	$\mu = \eta^2$

Exm

Example 11.1 (Matching outcomes to families and links)**.** Each of the following outcomes calls for a different choice of family and link:

Outcome	Data type	Family	Link
systolic blood pressure (mmHg)	continuous, unbounded in practice	Gaussian	identity
whether a participant develops diabetes during follow-up	binary	binomial	logit
number of chronic medical conditions (Example 10.2)	count	Poisson	log
hospital length of stay (days), measured continuously	positive and right-skewed	gamma	log

For the last outcome, the canonical inverse link of the gamma family would also be allowed, but the log link keeps the fitted means positive for every value of the linear predictor and makes $\exp\{\beta_j\}$ a ratio of means, just as in Example 10.2.

Exercises

Exercise 11.1 (Identifying GLM components)**.** (adapted from Dobson and Barnett (2018), Chapter 4, and Dunn and Smyth (2018), Chapter 5)
For each of the following scenarios, identify the three components of the GLM: (1) the outcome distribution, (2) the link function $g(\cdot)$, and (3) the linear predictor $\eta = \tilde{x} \cdot \tilde{\beta}$.
(a) Modeling the probability of a binary outcome (disease/no disease) as a function of continuous predictors, using the logit link.
(b) Modeling count data (number of events per person-year) as a function of continuous predictors, using the log link.
(c) Modeling a continuous, strictly positive outcome using a gamma distribution and log link.

Solution

Solution. **(a)** Logistic regression:

1. **Outcome distribution**: $Y_i \mid \tilde{x}_i \sim_{\perp\!\!\!\perp} \text{Bernoulli}(\pi_i)$
2. **Link function**: logit link, $g(\pi) = \log\{\pi/(1-\pi)\}$
3. **Linear predictor**: $\eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$

The mean function (inverse link) is: $\pi_i = \text{expit}(\eta_i) = \frac{e^{\eta_i}}{1+e^{\eta_i}}$.

(b) Poisson (log-linear) regression:

1. **Outcome distribution**: $Y_i \mid \tilde{x}_i \sim_{\perp\!\!\!\perp} \text{Pois}(\mu_i)$
2. **Link function**: log link, $g(\mu) = \log\{\mu\}$
3. **Linear predictor**: $\eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$

The mean function (inverse link) is: $\mu_i = e^{\eta_i}$.

(c) Gamma regression with log link:

1. **Outcome distribution**: $Y_i \mid \tilde{x}_i \sim_{\perp\!\!\!\perp} \text{Gamma}(\text{mean} = \mu_i)$
2. **Link function**: log link, $g(\mu) = \log\{\mu\}$
3. **Linear predictor**: $\eta_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}$

The mean function is: $\mu_i = e^{\eta_i}$.

Exercise 11.2 (Canonical link functions)**.** (adapted from Dobson and Barnett (2018), Chapter 3, and Dunn and Smyth (2018), Chapter 5)

The *canonical link* for a distribution in the exponential family is the link $g(\cdot)$ such that $g(\mu) = \theta$, where θ is the natural parameter (see Canonical links[a]).

For each listed distribution, state the canonical link function and the corresponding variance function $V(\mu)$, where $\text{Var}(Y) = \phi\, V(\mu)$ and ϕ is the dispersion parameter:

(a) $Y \sim \text{Pois}(\mu)$

(b) $Y \sim \text{Bernoulli}(\pi)$

(c) $Y \sim \text{N}(\mu, \sigma^2)$

(d) $Y \sim \text{Gamma}(\mu, \phi)$, with mean μ and shape $1/\phi$

[a] Intro-to-GLMs.qmd#sec-canonical-link

Solution

Solution.

Distribution	Canonical link	Variance function $V(\mu)$	$\text{Var}(Y) = \phi V(\mu)$
(a) Poisson	$g(\mu) = \log\{\mu\}$	$V(\mu) = \mu$	$\phi = 1$ (fixed)
(b) Bernoulli	$g(\pi) = \log\{\frac{\pi}{1-\pi}\}$ (logit)	$V(\pi) = \pi(1-\pi)$	$\phi = 1$ (fixed)
(c) Normal	$g(\mu) = \mu$ (identity)	$V(\mu) = 1$	$\text{Var}(Y) = \sigma^2$
(d) Gamma	$g(\mu) = -1/\mu$ (inverse)	$V(\mu) = \mu^2$	$\text{Var}(Y) = \phi\mu^2$

Notes:

- For the Poisson distribution, the variance equals the mean. Overdispersion occurs when the empirical variance exceeds the mean.
- For the Bernoulli distribution, the variance is maximized at $\pi = 0.5$.
- For the Normal distribution, the variance does not depend on the mean; the dispersion parameter $\phi = \sigma^2$.
- For the Gamma distribution, the coefficient of variation $\mathrm{SD}(Y)/\,\mathrm{E}[Y]$ is constant.
- The sign of the Gamma canonical link is a convention: $\theta = -1/\mu$ in the density's natural parameterization, while R's `Gamma()` family uses $1/\mu$ as its default inverse link. Either way, the linear predictor can produce a negative μ.

The log link (not the canonical inverse link) is often used for Gamma regression in practice, because it ensures $\mu > 0$ and gives interpretable multiplicative effects.

References

Dalgaard, Peter. 2008. *Introductory Statistics with r*. New York, NY: Springer New York. https://link.springer.com/book/10.1007/978-0-387-79054-1.

Dobson, Annette J, and Adrian G Barnett. 2018. *An Introduction to Generalized Linear Models*. 4th ed. CRC press. https://doi.org/10.1201/9781315182780.

Dunn, Peter K, and Gordon K Smyth. 2018. *Generalized Linear Models with Examples in R*. Vol. 53. Springer. https://link.springer.com/book/10.1007/978-1-4419-0118-7.

Hernán, Miguel A., and James M. Robins. 2020. *Causal Inference: What If*. Chapman & Hall/CRC. https://www.hsph.harvard.edu/miguel-hernan/causal-inference-book/.

James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani. 2021. *An Introduction to Statistical Learning: With Applications in R*. 2nd ed. Springer. https://doi.org/10.1007/978-1-0716-1418-1.

Kleinbaum, David G, Lawrence L Kupper, Azhar Nizam, K Muller, and ES Rosenberg. 2014. *Applied Regression Analysis and Other Multivariable Methods*. 5th ed. Cengage Learning. https://www.cengage.com/c/applied-regression-analysis-and-other-multivariable-methods-5e-kleinbaum/9781285051086/.

Neyman, Jerzy. 1990. "On the Application of Probability Theory to Agricultural Experiments. Essay on Principles. Section 9." *Statistical Science* 5 (4): 465–72. https://doi.org/10.1214/ss/1177012031.

Rubin, Donald B. 1974. "Estimating Causal Effects of Treatments in Randomized and Nonrandomized Studies." *Journal of Educational Psychology* 66 (5): 688–701. https://doi.org/10.1037/h0037350.

Vittinghoff, Eric, David V Glidden, Stephen C Shiboski, and Charles E McCulloch. 2012. *Regression Methods in Biostatistics: Linear, Logistic, Survival, and Repeated Measures Models*. 2nd ed. Springer. https://doi.org/10.1007/978-1-4614-1353-0.

Wickham, Hadley, Mine Çetinkaya-Rundel, and Garrett Grolemund. 2023. *R for Data Science*. " O'Reilly Media, Inc." https://r4ds.hadley.nz/.